

Bureaucratic Incentives and Data Production: Evidence from Social Registries

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1. Lots of information about lots of households → **effort intensive**
 - Require routine updating (in contrast to, e.g., vital statistics)
2. **Targeted population** less likely to be otherwise “legible” to state
 - Esp. in settings with high levels of informality
3. Information inputs → **monetary benefits**

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Variation in quality, comprehensiveness of registries at all levels of capacity:

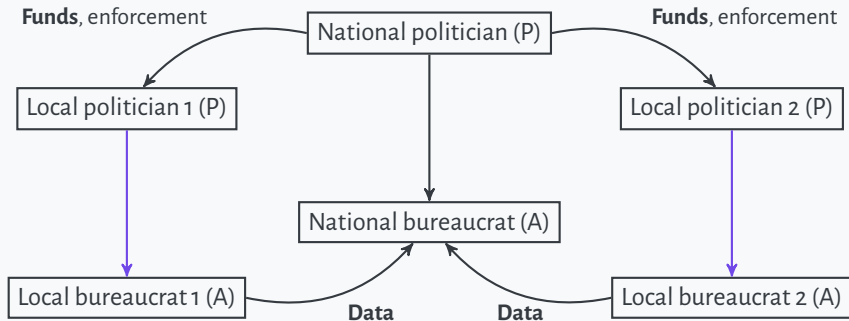
- Example: No national registry of SNAP recipients in the US

Intergovernmental relations and register production

Many administrative data, including social registers, are produced for **central government** by bureaucrats working in **state or local** governments.

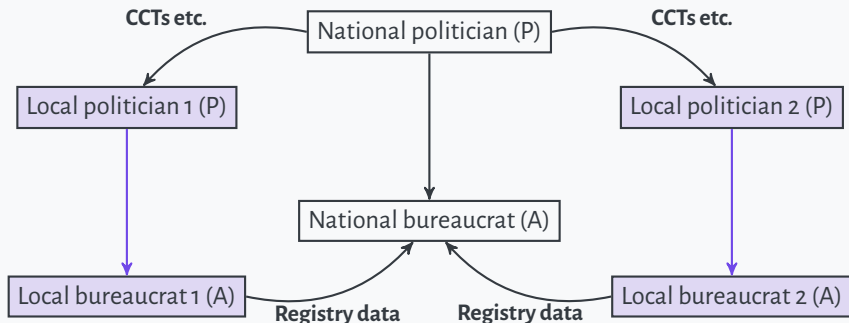
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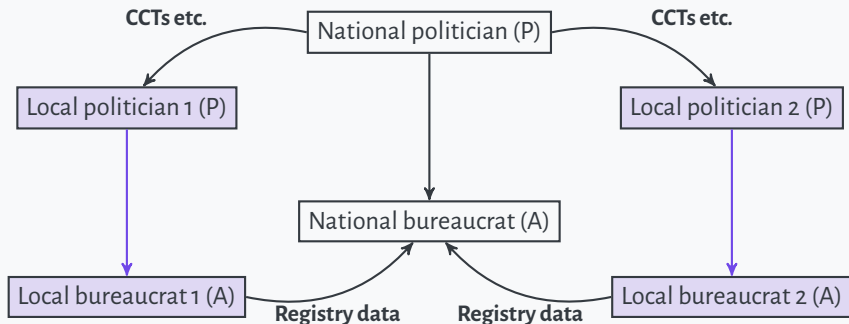
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This paper: Local politicians use **personnel policy** to shape local bureaucrats' **reporting behavior**, which in turn affects constituents' **access to social programs**.

THEORY

Overview

A model in which:

- A politician chooses whether to **select** new bureaucrat or **retain** predecessor's register administrator.
- Politician commits to level of **oversight**.
- Bureaucrat exerts **effort** to register households.
 - Determines share of households scored, a prerequisite for accessing means-tested social programs.
 - Scores depend on bureaucratic knowledge, preferences.

Preferences

Politician wants to draw transfers to jurisdiction → **maximize qualification** for social programs: **true positive** and **false positive** qualifications

Two (private) types of bureaucrat vary in preferences over policy outputs

1. **Technocrats**: Value accuracy: **true positive** qualifications, **true negative** denials.
2. **Loyalists**: Value $\uparrow$ qualification like politician, i.e., **true** and **false positives**

For both types of bureaucrat:

- Effort is equally costly
- Baseline knowledge of a new bureaucrat is the same → ability to score households in alignment with preferences

Core tradeoff

By **appointing a new bureaucrat**, a politician:

- ↑ probability of a loyalist
- ↓ bureaucratic knowledge

By **retaining the old bureaucrat**, a politician:

- ↓ probability of a loyalist
- ↑ bureaucratic knowledge

Core tradeoff

By **appointing a new bureaucrat**, a politician:

- $\uparrow$ probability of a loyalist
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By **retaining the old bureaucrat**, a politician:

- $\downarrow$ probability of a loyalist
- $\uparrow$ bureaucratic knowledge

Politician's monitoring intensity and **bureaucrat's effort** feed into politician's selection of the bureaucrat in equilibrium.

Key theoretical results

All else equal, loyalists exert **more effort** than technocrats.

- Errors of inclusion require effort but correct denials do not necessarily.

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All else equal, loyalists exert **more effort** than technocrats.

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All else equal, politician **monitors more** when there are more loyalists in the pool (of possible bureaucrats)

- Loyalists are more common among new appointees.
- Implies that selection of new bureaucrat and monitoring are **strategic complements**

Equilibrium implications

In settings with sufficiently large **changes in the pool** of new appointees versus potential holdovers:

1. Appointed bureaucrats are monitored more.
2. Appointed bureaucrats exert more effort.
3. Appointed bureaucrats classify households as poorer to increase social program eligibility.
4. Incremental gains in knowledge over a bureaucrat's tenure (fixing monitoring) lead to increases in the share of eligible classifications.

CONTEXTS

Overview: Brazil and Colombia

Commonality: Data production process (i.e., actors, institutions)

- National government designs register, social programs (e.g., CCTs)
- **Local governments** to maintain social register data
 - Local mayors responsible for hiring, overseeing bureaucrats who maintain registers.
 - Broad discretion to transfer bureaucrats, some ability to fire bureaucrats.

Difference: **Design** of the social registries, means-tested social policies

- Means-testing measure: income (Brazil), $\approx$ index of assets (Colombia)
- More fragmented set of social policies in Colombia

Bureaucratic tenure in local government, job

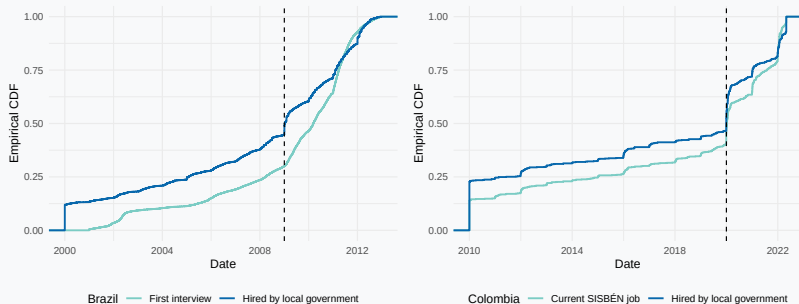


Figure: Tenure of bureaucrats (a) in local governments and (b) as agents running/administering social registries in Brazil (left) and Colombia (right). Tenures are censored to 12 years.

Brazilian, Colombian bureaucrats are largely similar on observables (age, gender, education)

Registries: CadUnico and SISBÉN-IV

	Brazil	Colombia
Registry	Cadastro Único para Programas Sociais (CadUnico)	Sistema de Identificación de Potenciales Beneficiarios de Programas Sociales (SISBÉN-IV)
Created	2002	1995 (1997)
Households in registry	16.7m (in 2012)	9.2m (in 2022)
Main CCT	Bolsa Familia	(Más) Familias en Acción*
CCT budget/GDP	0.43% (in 2012)	0.17% (in 2021)
CCT households	13.9m (in 2012)	3.1m (in 2022)
Eligibility based on	(Self)-reported income	Unknown mapping: survey data → vulnerability categories

Table: CCT budget, enrollment data from CEPAL. *Colombia has a wider variety of social programs for which SISBÉN determines eligibility.

RESEARCH DESIGN

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Examine **bureaucratic employment** → bureaucrats' **social registry reports**.

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Different research designs → different empirical targets

- Distinct sources of data on bureaucratic employment: survey (Colombia) and employer-employee data (Brazil).
- Different design of social registries: outcomes of interest.

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Approach:

1. Comparison of directional effects where “related” estimates are feasible.
2. Analysis in a single context where warranted.

Personnel data

Brazil: Administrative employer-employee data (RAIS)

- Identify bureaucrats by matching ID number of interviewer from CadUnico registry to employer-employee data $\rightarrow n = 33,671$ unique bureaucrats in 5,385 municipalities
- RAIS includes individual-level covariates, features of employment (i.e., tenure)

Colombia: Original survey of SISBÉN administrators (in 2022)

- Survey one administrator per locality, identified in 1,074/1,102 municipalities by FOI requests to national gov't
- Response rate: 754/1,074 (70.2%)

Contrast: Bureaucratic appointment

Contrast: Compare the behavior of register administrators

Appointed by current mayor vs. **Retained** by current mayor

In Colombia: Coded from survey-reported start of tenure as registry administrator (pre- vs. post-Jan 1, 2020)

- 54.9% of 750 surveyed bureaucrats appointed by mayor

In Brazil: Coded from RAIS based on hire date of registry interviewer (pre- vs post-Jan 1, 2009)

- 58.6% of 33,671 bureaucrats appointed by mayor during current term
- 6.25 interviewers/municipality on average (median of 4)
- Within municipality variation in 3,294 municipalities (62%)

Outcomes data: Registries

Attribute	CadUnico (Brazil)	SISBÉN-IV (Colombia)
	PANEL A: RAW DATA	
Cross-section Time Sample	Household December 2012 Universe of households	Household or individual Annual from 2021-2022 Random sample by municipality-zone

Table: Raw administrative social register data and aggregate units of analysis.

Outcomes data: Registries

Attribute	CadUnico (Brazil)	SISBÉN-IV (Colombia)
	PANEL A: RAW DATA	
Cross-section	Household	Household or individual
Time	December 2012	Annual from 2021-2022
Sample	Universe of households	Random sample by municipality-zone
	PANEL B: PROCESSED DATA	
Unit of analysis	Bureaucrat	Municipality-zone year
Years analyzed	2009-2012	2021, 2022, (2023)
Total municipalities	5,568	1,102
Municipalities in sample	5,385	754
Sample restrictions	Municipality present in CadUnico ($n = 5554$), municipal government present in RAIS ($n = 5,554$), in which any interviewer from 2009-2012 term is located in RAIS data ($n = 5,385$).	SISBÉN administrator answered survey ($n = 754$).

Table: Raw administrative social register data and aggregate units of analysis.

Outcome variables

Outcome	Brazil	Colombia
OUTCOMES FROM SURVEY		
Monitoring		- Oversight frequency - Goals set by supervisor - Perceived freedom
OUTCOMES FROM CADUNICO AND SISBÉN DATA		
Effort	- # of interviews conducted by bureaucrat	- Per-capita rate of SISBÉN enrollment within municipal-zone
Scores	- % HHs under BF unconditional transfer threshold - % HHs under BF conditional transfer threshold - % HHs at BF unconditional threshold - % HHs at BF conditional threshold	- Distribution of scores by ranked poverty category
Program receipt	% HHs enrolled in BF	

Table: Outcomes variables of interest from survey and social registry data

RESULTS

Implication #1: Selection and Monitoring

Remark

For a sufficiently large change in the pool of holdovers vs. new appointees, politicians impose stronger oversight among new appointees.

Implies $\beta_1 > 0$ in the following OLS specification:

$$\text{Oversight}_m = \beta_1 \text{Appointed}_m + \beta_2 \text{Contractor}_m + \gamma \mathbf{X}_m + \varepsilon_i$$

Covariates:

- Contractor_m : indicator for contractor (not civil servant)
- $\mathbf{X}_m$: Bureaucrat characteristics (age bins, education level, gender) and municipal characteristics (population quintile, municipal category)

Mayors monitor their new appointees more

	Oversight frequency		Oversight by mayor		Goals set		(Lack of) freedom	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Appointed by mayor	0.441*** (0.096)	0.418*** (0.109)	0.052 (0.036)	-0.011 (0.040)	0.085* (0.035)	0.117** (0.039)	0.071 (0.083)	-0.021 (0.095)
Observations	738	734	750	745	741	737	739	735
Bureaucrat characteristics	✓		✓		✓		✓	
DV scale	{0, ..., 5}		{0, 1}		{0, 1}		{1, ..., 5}	
DV mean, std. dev.	3.133 (1.249)		0.367 (0.482)		0.703 (0.457)		2.333 (1.101)	

+p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appointed mayors report ↑ oversight frequency, goal setting

- Even though their preferences are more closely aligned to mayor's
- Mechanism: ↑ marginal return to monitoring → ↑ monitoring
- Robust to adjustment for civil servant/contractor employment status
- Not driven by time on the job

Implication #2: Bureaucratic effort

Remark

For a sufficiently large change in the pool of holdovers vs. new appointees, new appointees exert greater effort in expectation.

Implies $\beta_1 > 0$ in the following OLS specifications.

Colombia: z indexes municipal zone

$$\Delta \text{Share enrolled}_{m(z), 2022-2021} = \beta_1 \text{Appointed}_m + \beta_2 \text{Contractor}_m + \kappa \mathbf{X}_m + \epsilon_{mz}$$

Brazil: i indexes bureaucrats and m indexes municipalities.

$$\ln(\text{Interviews}_{im}) = \beta_1 \text{Appointed}_i + \beta_2 \text{Contractor}_i + \psi_m + \kappa \mathbf{X}_i + \epsilon_{im}$$

Mayoral appointees register more households in Colombia

	Δ Share of households enrolled, 2021-2022					
	(1)	(2)	(3)	(4)	(5)	(6)
ALL HOUSEHOLDS IN MUNICIPALITY						
Appointed by mayor	0.022** (0.008)	0.039*** (0.011)	0.031* (0.012)	0.018* (0.009)	0.017+ (0.009)	0.016+ (0.008)
DV mean	0.156	0.156	0.156	0.156	0.156	0.156
Observations	747	747	747	747	747	747
Contractor		✓	✓	✓	✓	✓
Bureaucrat covariates			✓	✓	✓	✓
Municipal covariates				✓	✓	✓
Share of households enrolled, 2021					✓	✓
Δ Enrollment, SISBÉN-III to IV						✓
Missingness indicators		✓	✓	✓	✓	✓

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Full Colombia results

Baseline enrollment in Colombia

Mayoral appointees register more households in Brazil

	Log Households Interviewed			
	(1)	(2)	(3)	(4)
PANEL A: HOUSEHOLDS INTERVIEWED IN 2011				
Appointed by mayor	0.120** (0.041)	0.122** (0.042)	0.097* (0.042)	0.053* (0.022)
Observations	25316	25316	25316	25316
PANEL B: HOUSEHOLDS INTERVIEWED IN 2012				
Appointed by mayor	0.203*** (0.041)	0.205*** (0.042)	0.189*** (0.041)	0.168*** (0.023)
Observations	25680	25680	25680	25680
Municipal FE	✓	✓	✓	✓
Contractor Indicator		✓	✓	✓
Bureaucrat Characteristics			✓	✓
Interview experience FE				✓

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table: Bureaucratic appointment and output in Brazil. Standard errors are clustered at the municipality level.

Similar results using all interviews in cross-section of CadUnico. [Full Brazil results](#)

Implication #3: Scores of registered households

Remark

For a sufficiently large change in the pool of holdovers vs. new appointees, new appointees report more households below the eligibility threshold than retained bureaucrats.

In Colombia, outcome is the change in the CMF of ranked poverty categories, c :

$$\sum_1^c \text{Share of enrolled households}_{m(z), 2022} - \sum_1^c \text{Share of enrolled households}_{m(z), 2021}$$

In Brazil, outcomes are bureaucrat-level average scores or outcomes.

Use same estimators as for the enrollment specification.

New appointees increase the households in poverty: Colombia

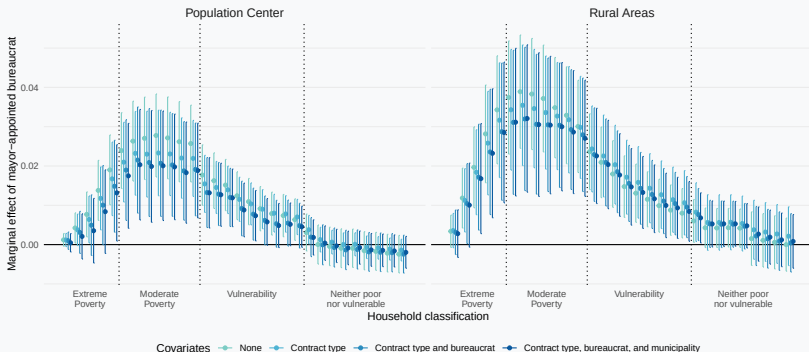


Figure: Marginal effects of a mayor-appointed bureaucrat on differences in the 2022 and 2021 cumulative mass functions of household classifications.

In Colombia, mayoral appointees:

- Shift more households to poorer categories between 2021 and 2022
- Differences cannot be explained by (targeted) enrollment alone

Reclassification

New appointees increase the households in poverty, Brazil

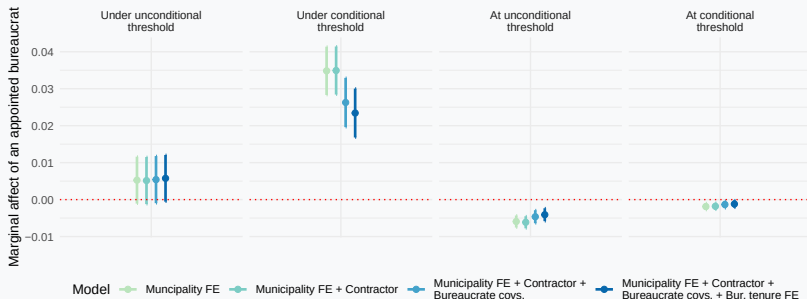


Figure: Estimates of the marginal effect of being appointed by the current mayor on income classification measures. Outcomes are the share of bureaucrat's interviews with a given label.

In Brazil, mayoral appointees:

- ↑ Share of households below Bolsa Familia thresholds
- ↓ Bunching at threshold → consistent with increased effort

Implications for Bolsa Familia Enrollment

To recap, relative to holdovers, new appointees:

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1. Exert more effort to score more households
2. Conditional on interviewing, record more BF-eligible households

	Share of interviewed households in Bolsa Familia roll			
	(1)	(2)	(3)	(4)
Appointed by mayor	0.048*** (0.004)	0.049*** (0.004)	0.035*** (0.004)	0.029*** (0.004)
Observations	33,671	33,671	33,671	33,671
DV Mean (Std. Dev.)	0.530 (0.279)	0.530 (0.279)	0.530 (0.279)	0.530 (0.279)
Within-muni. DV std. dev.	0.217	0.217	0.217	0.217
Municipal FE	✓	✓	✓	✓
Contractor Indicator		✓	✓	✓
Bureaucrat Characteristics			✓	✓
Interview experience FE				✓

Full results

Broader implications

DISCUSSION

On the reform of social registries

Two approaches to improve the accuracy of social registry data:

- **Brazil:** Incremental increases in thresholds; Top-down audits of individual households through cross-checking national databases
- **Colombia:** Large-scale reform of survey, index construction; Re-surveying households

Downside of large-scale reform: ↓ knowledge of *all* sitting bureaucrats

- ↓ incentive to retain bureaucrats
- But this reduces share of technocrats among register administrators → bad for national government!
- Incremental reforms may be preferable with respect to post-reform registry production.

On means testing social programs

Should social programs like CCTs be means-tested or universal?

Ignoring costs of data collection, cost of benefits in means-tested programs approach those of universal programs when:

- Subset of population that should qualify is large
- Type-I errors (of inclusion) are more frequent → More likely in patronage-heavy environments with many **loyalists** among administrators

Universal programs may be comparatively more attractive to the national government when local bureaucrats are less insulated from local politicians.

Take Aways

Politician's personnel choices leave “footprints” in social registries

Moral hazard problem of local bureaucrats insulate (to varying degrees) means testing from local politicians

- ... but politicians push back through staffing choices

We need more studies of how selection + oversight work in tandem

- New mechanism to explain observed complementarity: $\uparrow$ marginal returns to monitoring among new appointees.
- Likely in settings where policy or public service motivation motivate different levels of effort between appointed and retained bureaucrats.

THANK YOU!

Comments welcome!
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